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Classical Field Theory (CFT) and Principle of Locality

Preparation:

1. Goal of QFT
 - ① concept, formalism and techniques for QFT develop
 - ② understand QED
 - ③ 3/4 fundamental interaction in the nature
 - ④ particle physics + condensed matter + statistical physics $\rightarrow$ QFT is a universal language
2. Design of this class:
 - ① QFT I: fundamental concepts
 - ② QFT II: Technical development.
3. Outline on the website: (don't treat it too literally).
4. Three pieces of advice on QFT:
 - ① QM + ∞ number Degree of freedom (DOF)
 $\rightarrow$ fill the gap in understanding QM.
 - ② QFT deals with formalism. $\rightarrow$ concrete physical problems
 - ③ Intuition is built through experiences.
5. Notation:



Lecture 1:

Chap 1: Why QFT?

1.1 Classical Field Theory (CFT)

- principle of locality.

Newtonian mechanics: action at a distance.

Faraday (1830): all points in space participate

the physical process, effects propagate

from pt to neighbouring pts

Fields: the mathematical device (vehicular)

that the principle of locality is at work

Associate with each pt in the space a

dynamical variable:

e.x 1: electric field $\vec{E}(\vec{x}, t) \equiv \vec{E}(\vec{x}, t)$

magnetic field $\vec{B}(\vec{x}, t) \equiv \vec{B}(\vec{x}, t)$

$\vec{x}$: labels

Maxwell's eqn

$$\begin{cases} \nabla \cdot \vec{B} = 0 \\ \nabla \cdot \vec{E} = 4\pi\rho \\ \frac{\partial \vec{B}}{\partial t} = -\nabla \times \vec{E} \\ \nabla \times \vec{B} = 4\pi\vec{J} + \frac{\partial \vec{E}}{\partial t} \end{cases}$$

Local eqns: contain only $\vec{E}$ and $\vec{B}$

at same pt with finite # of derivatives

e.x. 2: Einstein's gravity

$g_{\mu\nu}(\vec{x}, t)$





Types of fields:

Scalar field: $\phi(\vec{x}, t)$

vector field: $\vec{E}(\vec{x}, t), \vec{B}(\vec{x}, t).$

$A_\mu(x^\nu)$. (vector potential)

tensor field: $f_{\mu\nu}(x^\mu)$

spinor field: $\psi_\alpha(\vec{x}, t)$

Conventions: $\mu, \nu = 0, 1, 2, 3$

$$x^\mu = (ct, \vec{x}) = (ct, x^i)$$

$$\partial_\mu = (\frac{1}{c}\partial_t, \nabla)$$

$$A_\mu = (A_0, \vec{A}) = (A_0, A_i)$$

$$c=1, \hbar=1.$$

Action principle:

In classical mechanics:

$$S = \int dt L(x, \dot{x}, t)$$

$$p = \frac{\partial L}{\partial \dot{x}}, \quad H = p\dot{x} - L \quad (\text{Legendre transf})$$

EOM: extremize $S[x(t)]$

From principle of locality:

L is significantly constrained. It must have the

form:

$$L = \int d^3x \mathcal{L}(\phi_a, \partial_\mu \phi_a, \partial_\mu \phi)$$



where: ① $\phi_a(\vec{x}, t)$ is general fields
 $\uparrow$
 a labels diff fields

② $\mathcal{L}$ is a function of ϕ_a , and $\partial_t \phi_a$, $\partial_i \phi_a$.
 i.e. $\mathcal{L}$ only depends on the value of ϕ_a and
 its derivatives at single pt.

$\mathcal{L}$ is called: Lagrangian density

Remarks: (i) principle of locality: does not allow

$$\mathcal{L} = \dots + \int d^3x d^3y \phi(\vec{x}, t) \phi(\vec{y}, t)$$

(ii) only allow: $\partial_t \phi \Rightarrow$ eom contains 2 time derivatives.

Canonical momentum density:

$$\pi_a(\vec{x}, t) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_a(\vec{x}, t)}$$

$$\dot{\phi}_a = \partial_t \phi_a(\vec{x}, t)$$

Hamiltonian density:

$$\mathcal{H}(\vec{x}, t) = \dot{\phi}_a \pi_a - \mathcal{L}$$

(a is summed, Einstein Convention)

Hamiltonian:

$$H = \int d^3x \mathcal{H}$$

EOM: $\delta S = 0$

$$S = \int d^4x \mathcal{L}(\phi_a, \partial_\mu \phi_a) = \int dt \mathcal{L}$$

$$0 = \delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \delta (\partial_\mu \phi_a) \right]$$

integration by part.

$$= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) \right] \delta \phi_a + \text{bdry terms}$$

choose $\delta \phi_a$ such that bdry terms vanish.

(repeated terms are summed).

(4)





$$\Rightarrow \frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) = 0 \text{ general EoM for cFT}$$

From now on:

1) translationally invariant:

2) Lorentz invariant

E.x. 1) Maxwell's theory:

dynamical variables: $A_\mu(x)$.

Field strength: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

$$S_{EM} = \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right).$$

2) Einstein gravity:

$$S = \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} R.$$

3) Scalar field theory:

consider a real value scalar field $\phi(x)$.

$$S = \int d^4x \left[-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) \right]$$

$V(\phi)$ = function of ϕ

translation symmetry $\Rightarrow$ all parameters in $V(\phi)$ must be constants



$$\pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}(x).$$

$$(\partial_\mu \phi \partial^\mu \phi = -\dot{\phi}^2 + (\nabla \phi)^2)$$

$$\mathcal{H} = \pi \dot{\phi} - \mathcal{L} = \frac{\pi^2}{2} + \frac{1}{2}(\nabla \phi)^2 + V(\phi).$$

$$\text{EOM: } \partial^2 \phi - \frac{\partial V}{\partial \phi} = 0 \quad (\partial^2 = \partial_\mu \partial^\mu = -\partial_t^2 + \nabla^2).$$

① Simplest case: $V(\phi) = f\phi$. (f is const from principle of locality).

$$\Rightarrow \text{EOM: } \partial^2 \phi = f \quad "f \text{ is external force}"$$

$$\textcircled{2} V(\phi) = \frac{1}{2} m^2 \phi^2.$$

$$\partial^2 \phi - m^2 \phi = 0 \quad (\text{Klein-Gordon eqn}).$$

